First Midterm Exam

Name: _____

Please circle your section number!

Section 1	Section 2	Section 3
Ander Aguirre (MWF 9:55 AM)	Sarah Tammen (TR 9:30 AM)	Nicholas Derr (MWF 9:50 AM)

- **There are 5 problems on the exam, some of them have multiple parts.**
- Read the problems carefully and budget your time wisely.
- If you are running out of time, please at least say *how* you would solve a given problem, preferably using notation, even if you do not have the time to do so. This may earn you partial credit.
- No calculators or other electronic devices, please. **Turn off your phone.**
- You do not have to carry out complicated numerical computations, but you should simplify your answer if it is possible with reasonable effort. In particular, geometric series must be evaluated.
- Please present your solutions in a clear manner. **Justify your steps.** Numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
- You can use the empty pages for additional space for writing if necessary.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
Total:	60	

1. Suppose we perform an experiment in which we sample 3 balls without replacement from an urn filled with red, white, and green balls. Suppose that there are twenty balls total, and suppose for convenience that they are numbered; specifically, suppose that the red balls are numbered 1-10, that the white balls are numbered 11-15, and that the green balls are numbered 16-20. Suppose we write down the numbers on the balls in the order we draw them.

   (a) (4 points) What is the sample space of this experiment? If we let our set of events consist of all outcomes of the sample space, what is an appropriate probability measure?

   Solution:
   Since we are sampling with order, our sample space should consist of ordered triples of integers. Since we sample without replacement, all three coordinates of each triple should be distinct. Thus, we take

   $$\Omega = \{(x_1, x_2, x_3) : x_1, x_2, x_3 \in \{1, 2, \ldots, 20\} \text{ with } x_i \neq x_j \text{ if } i \neq j\}.$$

   Each triple has an equal probability of being chosen. Thus for any triple (x_1, x_2, x_3), we have

   $$P((x_1, x_2, x_3)) = \frac{1}{\#\Omega} = \frac{1}{20 \cdot 19 \cdot 18},$$

   and any event A has probability

   $$P(A) = \frac{\#A}{\#\Omega} = \frac{\#A}{20 \cdot 19 \cdot 18}.$$

   (b) (6 points) Determine the probability of each of the following events. You may leave your answers unsimplified.

      i. The event that all three balls in the sample are red

      Solution:
      The probability of interest is

      $$\frac{10 \cdot 9 \cdot 8}{20 \cdot 19 \cdot 18}.$$

      One way to see this is as follows: if we let R_i be the event that ball i is red for $i = 1, 2, 3$, then we have that

      $$P(R_1 R_2 R_3) = P(R_1)P(R_2|R_1)P(R_3|R_1 R_2) = \frac{10}{20}\left(\frac{9}{19}\right)\left(\frac{8}{18}\right).$$

      Note, however, that we gave full credit for a correct answer with no justification. Note also that the above expression is equivalent to

      $$\frac{\binom{10}{3}}{\binom{20}{3}},$$

      so we gave full credit for this as well.

      ii. The event that none of the balls in the sample are red

      Solution:
      The probability of interest is again

      $$\frac{10 \cdot 9 \cdot 8}{20 \cdot 19 \cdot 18}.$$

If we borrow the notation from part (i), then we can see this as follows:

$$P(R_1^c R_2^c R_3^c) = P(R_1^c)P(R_2^c|R_1^c)P(R_3^c|R_1^c R_2^c) = \frac{10}{20}\left(\frac{9}{19}\right)\left(\frac{8}{18}\right).$$

iii. The event that the sample contains balls of at least two different colors

Solution:
We have that

$$\begin{aligned} P(\{\text{at least two different colors}\}) &= 1 - P(\{\text{all balls are the same color}\}) \\ &= 1 - (P(\{\text{all balls red}\}) - P(\{\text{all balls green}\}) - P(\{\text{all balls white}\})) \\ &= 1 - \frac{10 \cdot 9 \cdot 8}{20 \cdot 19 \cdot 18} - 2\left(\frac{5 \cdot 4 \cdot 3}{20 \cdot 19 \cdot 18}\right). \end{aligned}$$

2. Suppose we are given a pair of fair tetrahedral dice. (A tetrahedral die is a triangular pyramid with the faces numbered 1-4.) Let X_1 be the first roll, let X_2 be the second roll, and let S be the sum of the rolls.

(a) (3 points) Determine the probability mass function of S.

Solution:
The possible values of S are $2, 3, 4, 5, 6, 7, 8$, and S has probability mass function

$$p_S(k) = \begin{cases} \frac{1}{16}, & k \in \{2, 8\}; \\ \frac{2}{16} = \frac{1}{8}, & k \in \{3, 7\}; \\ \frac{3}{16}, & k \in \{4, 6\}; \\ \frac{4}{16}, & k = 5. \end{cases}$$

One way to see this is to make a chart whose (i, j)-entry is $P(X_1 = i, X_2 = j)$ for $i, j \in \{1, 2, 3, 4\}$.

(b) (7 points) (2+5) Suppose we roll the dice repeatedly until the sum of the rolls is 4 or 6 and agree that I will win if we see a sum of 4 first but you will win if we see a sum of 6 first.

i. Let N be the number of rounds required to determine a winner, where a round consists of rolling both dice. What is the distribution of N? Identify the name of the distribution and all relevant parameters.

Solution:
We have that $N \sim \text{Geom}\left(\frac{3}{8}\right)$ so that N has probability mass function

$$p_N(k) = \left(\frac{5}{8}\right)^{k-1}\left(\frac{3}{8}\right), \qquad k \in \mathbb{N}.$$

ii. Determine, with justification, the probability that the probability that the game ends on or before the third round with a sum of six.
Solution:
Let A be the event that the game ends with a sum of 6. For each $i \in \mathbb{N}$, let B_i be the event that $N = i$. We are interested in computing $P(A \cap \{N \leq 3\})$. Since the event that $N \leq 3$ is the disjoint union $B_1 \cup B_2 \cup B_3$), we have that

$$\begin{aligned} P(A \cap \{N \leq 3\}) &= P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3) \\ &= P(A \cap B_1) + P(A|B_1^c)P(B_1^c) + P(A|B_1^c B_2^c)P(B_1^c B_2^c) \\ &= \frac{3}{16} + \frac{3}{16}\left(\frac{5}{8}\right) + \frac{3}{16}\left(\frac{5}{8}\right)^2 \\ & \frac{3}{16}\left(1 + \frac{5}{8} + \left(\frac{5}{8}\right)^2\right). \end{aligned}$$

3. Throughout this question, let A, B, C be three *mutually independent* events in a probability space with $P(A) = P(B) = P(C) = p$.

(a) (3 points) Show that $A \cap C$ is independent of B^c

Solution:

By mutual independence:

$$P((A \cap B^c) \cap C) = P(A \cap B^c \cap C) = P(AC) - P(ABC) = p^2 - p^3$$

Also by mutual independence:

$$P(A \cap B^c)P(C) = [P(A) - P(AB)]P(C) = p^2 - p^3$$

(b) (7 points) Compute, in terms of p, the following conditional probability

$$P(A \cap B^c \cap C \mid A \cup B^c \cup C).$$

Solution:

By mutual independence, Bayes theorem and inclusion-exclusion:

$$\begin{aligned} P(A \cap B^c \cap C \mid A \cup B^c \cup C) &= \frac{P(A)P(B^c)P(C)}{P(A) + P(B) + P(C) - P(AB^c) - P(B^cC) - P(AC) + P(AB^cC)} \\ &= \frac{p^2 - p^3}{2p + (1-p) - 2p(1-p) - p^2 + p^2(1-p)} \end{aligned}$$

4. We follow a state election in two counties: Adams and Brown. Our candidate either wins the state (W) or loses the state (L). Before any county results are reported we estimate even odds:

$$P(W) = 0.5, \qquad P(L) = 0.5.$$

Denote the events that our candidate wins Adams by A and Brown by B. Historical data shows:

$$P(A \mid W) = 0.8, \qquad P(B \mid W) = 0.6,$$

$$P(A \mid L) = 0.3, \qquad P(B \mid L) = 0.4.$$

Assume county outcomes are *conditionally independent* given W or L.

(a) (4 points) Suppose Adams reports first and our candidate wins it. Compute $P(W \mid A)$.

Solution:

$$P(W|A) = \frac{P(A|W)P(W)}{P(A|W)P(W) + P(A|L)P(L)} = \frac{0.4}{0.4 + 0.15} = \frac{8}{11}$$

(b) (6 points) Suppose our candidate wins Adams but loses Brown. Compute $P(W \mid A \cap B^c)$.

Solution:

Once again using Bayes theorem as well as conditional independence:

$$\begin{aligned} P(W|AB^C) &= \frac{P(A|W)P(B^c|W)P(W)}{P(A|W)P(B^c|W)P(W) + P(A|L)P(B^c|L)P(L)} \\ &= \frac{(0.8)(0.4)(0.5)}{(0.8)(0.4)(0.5) + (0.3)(0.6)(0.5)} = \frac{16}{25} \end{aligned}$$

5. Consider a random variable Y with probability density function

$$P(Y = y) = \begin{cases} y, & y \in [\frac{1}{2}, a), \\ 0, & \text{else.} \end{cases}$$

(a) (2 points) Find a such that $p_Y(y)$ is a valid probability distribution function.

Solution:

We require that the probability density function $f_Y(y)$ integrates to 1 over all possible values, so

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} f_Y(y)dy = \int_{1/2}^{a} ydy \\ &= \frac{y^2}{2}\Big]_{1/2}^{a} = \frac{a^2}{2} - \frac{1}{8} \\ &\implies \frac{9}{8} = \frac{a^2}{2} \\ &\implies \frac{9}{4} = a^2 \\ &\implies \boxed{a = \frac{3}{2}.} \end{aligned}$$

(b) (3 points) Introduce a new discrete random variable $U = \lfloor 2Y \rfloor$, where the floor operator $z = \lfloor r \rfloor$ represents the largest integer $z \leq r$ for $r \in \mathbb{R}$. Calculate the probability mass distribution of U.

Solution:

For each possible value u of U, the probability $P(U = u)$ will satisfy

$$P(U = u) = \int_{\{y : \lfloor 2y \rfloor = u\}} f_Y(y)dy,$$

i.e. we need to determine which y-values will yield that value of U and integrate the probability density over that range to obtain a probability mass. In this case, $y \in [\frac{1}{2}, 1)$ will correspond to $U = 1$, and $y \in [1, \frac{3}{2})$ to $U = 2$. The probability of U taking any other value is 0. Thus,

$$P(U = 1) = \int_{1/2}^{1} ydy = \frac{1^2}{2} - \frac{\left(\frac{1}{2}\right)^2}{2} = \frac{1}{2} - \frac{1}{8} \implies \boxed{P(U = 1) = \frac{3}{8},}$$

$$P(U = 2) = \int_{1}^{3/2} ydy = \frac{\left(\frac{3}{2}\right)^2}{2} - \frac{1^2}{2} = \frac{9}{8} - \frac{4}{8} \implies \boxed{P(U = 2) = \frac{5}{8}.}$$

(c) (2 points) Find b such that $P(Y < b) = 1/2$.

Solution:

For the probability density function $f_Y(y)$, we must have

$$P(Y \leq b) = \int_{-\infty}^{b} f_Y(y)dy = \int_{1/2}^{b} y\ dy = \frac{b^2}{2} - \frac{1}{8}.$$

Note that we know $b < \frac{3}{2}$, allowing us to just use y (not the full multiple-case function) as the integrand. Setting this value equal to $\frac{1}{2}$ as requested,

$$\frac{1}{2} = \frac{b^2}{2} - \frac{1}{8} \implies \frac{5}{8} = \frac{b^2}{2} \implies \frac{5}{4} = b^2 \implies \boxed{b = \frac{\sqrt{5}}{2}.}$$

(d) (3 points) Consider the events $A = \{Y \in [5/8, 12/8]\}$ and $B = \{Y \in [1, 2]\}$. Are they independent?

Solution:

We can compute the probability of each event as well as their intersection:

$$P(A) = \int_{5/8}^{12/8} f_Y(y)dy = \int_{5/8}^{3/2} y\ dy = \frac{\left(\frac{3}{2}\right)^2}{2} - \frac{\left(\frac{5}{8}\right)^2}{2} = \frac{9}{8} - \frac{25}{128} = \frac{144}{128} - \frac{25}{128} = \frac{119}{128} < 1.$$

We could also just have noted that $P(A) < 1$, since the interval corresponding to A doesn't include all possible values of Y. We also have

$$P(B) = \int_{1}^{2} f_Y(y) = \int_{1}^{3/2} y\ dy = \frac{5}{8}.$$

Finally, note that $B \implies A$, since there is no probability of $Y \in [\frac{3}{2}, 2]$, and the rest of B's interval lies within that of A. Thus,

$$P(AB) = P(B) = \frac{5}{8},$$

and since $P(A) < 1$, we have $\boxed{P(AB) = P(B) \neq P(A)P(B),}$ indicating they are **not** independent.

6. Consider the following experiment: you roll a six-sided die until you obtain a prime number (2, 3, or 5). Then, you perform a number of coin flips equal to the number of times you rolled the die. We define the random variables N as the number of rolls/flips and T as the number of tails obtained.

(a) (2 points) For each $n \in \mathbb{N}$, let B_n be the event that $N = n$. For a given n, what is the conditional probability $P(T \geq 1|B_n)$?

Solution:

First, we note that the complement may be easier to calculate, since

$$P(T = 0|B_n) = 1 - P(T \geq 1|B_n).$$

If the event B_n occurs, then $N = n$ and $(T|B_n) \sim \text{Bin}\left(n, \frac{1}{2}\right)$, so

$$P(T = 0|B_n) = \binom{n}{0}\left(\frac{1}{2}\right)^n = 2^{-n}.$$

Thus,

$$\boxed{P(T \geq 1|B_n) = 1 - 2^{-n}.}$$

(b) (3 points) What is the unconditional probability $P(T \geq 1)$?

Solution:

To obtain the unconditional probability, we must sum over all possible n, i.e.

$$P(T \geq 1) = \sum_{n=1}^{\infty} P(T \geq 1|B_n)P(B_n).$$

We know that $B_n \sim \text{Geom}\left(\frac{1}{2}\right)$, since there is probability 1/2 of rolling a 2, 3 or 5. Thus, $P(B_n) = 2^{-n}$ ($n - 1$ factors of 1/2 for $n - 1$ heads and one more for the final tails), and

$$P(T \geq 1) = \sum_{n=1}^{\infty}(1 - 2^{-n})2^{-n} = \sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^n - \sum_{n=1}^{\infty}\left(\frac{1}{4}\right)^n.$$

We can evaluate the series as

$$\sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^n = \frac{1}{2}\sum_{n=0}\left(\frac{1}{2}\right)^n = \frac{1}{2}\frac{1}{1-\frac{1}{2}} = 1,$$

$$\sum_{n=1}^{\infty}\left(\frac{1}{4}\right)^n = \frac{1}{4}\sum_{n=0}\left(\frac{1}{4}\right)^n = \frac{1}{4}\frac{1}{1-\frac{1}{4}} = \frac{1}{3},$$

so we conclude

$$\boxed{P(T \geq 1) = 1 - \frac{1}{3} = \frac{2}{3}.}$$

(c) (5 points) Give all possible values k of T and find the probability mass function $P(T = k)$. It may be useful to note that for $0 \leq p < 1$,

$$\sum_{k=0}^{\infty} \binom{k}{i} p^k = \frac{p^i}{(1-p)^{i+1}},$$

using the convention that $\binom{k}{i} = 0$ for $i < 0$ and $i > k$.

Solution:

We expect $T \in \mathbb{N} \cup \{\infty\}$, albeit with $P(T = \infty) = 0$. For $T = k$ we will need to sum over all possible n, i.e.

$$P(T = k) = P(\{T = k\} \cup B_n) = P(T = k|B_n)P(B_n).$$

This corresponds to thinking about all of the possible N with which we could have obtained that k. We already know $P(B_n) = 2^{-n}$. Since we know $T|B_n$ is binomial with n trials with $p = \frac{1}{2}$, we have

$$P(T = k|B_n) = \binom{n}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{n-k} = \binom{n}{k} \left(\frac{1}{2}\right)^n.$$

Thus we have

$$P(T = k) = \sum_{n=1}^{\infty} \binom{n}{k} \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} \binom{n}{k} \left(\frac{1}{4}\right)^n.$$

Note that since $\binom{n}{k} = 0$ for $k > n$, the terms in the sum will vanish for $n = 0, 1, 2, \ldots, k-1$. Specifically, for $k > 0$, the $n = 0$ term vanishes and the sum from $n = 1$ is equal to the sum from $n = 0$ (corresponding to the provided expression.) For the $k = 0$ term, we must correct for the difference, and we write

$$P(T = k) = \begin{cases} \sum_{n=0}^{\infty} \binom{n}{k} \left(\frac{1}{4}\right)^n, & k \geq 1, \\ -1 + \sum_{n=0}^{\infty} \binom{n}{k} \left(\frac{1}{4}\right)^n, & k = 0. \end{cases}$$

Now, we note that when we sub in the provided relation (with $k = n$, $i = k$, $p = 1/4$) we get

$$\sum_{n=0}^{\infty} \binom{n}{k} \left(\frac{1}{4}\right)^n = \frac{\left(\frac{1}{4}\right)^k}{\left(1 - \frac{1}{4}\right)^{1+k}} = \frac{4}{3^{1+k}}.$$

This, then, is our answer for $P(T = k)$ with $k > 0$, but for $k = 0$ we correct to get

$$P(T = 0) = -1 + \frac{4}{3} = \frac{1}{3}.$$

Thus, ultimately we get

$$\boxed{P(T = k) = \begin{cases} 4/3^{1+k}, & k \geq 1, \\ 1/3, & k = 0. \end{cases}}$$

So the first few values are

k	0	1	2	3	$\cdots$
$P(T = k)$	$\frac{1}{3}$	$\frac{4}{9}$	$\frac{4}{27}$	$\frac{4}{81}$	$\cdots$

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List of named distributions

$X \sim \text{Bernoulli}(p)$, PMF: $p_X(0) = 1 - p, \quad p_X(1) = p$

$X \sim \text{Binomial}(n, p)$, PMF: $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \le k \le n$

$X \sim \text{Geometric}(p)$, PMF: $p_X(k) = p(1-p)^{k-1}, \quad k = 1, 2, \ldots$

$X \sim \text{Poisson}(\lambda)$, PMF: $p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \ldots$

$X \sim \text{Hypergeometric}(N, N_A, n)$, PMF: $p_X(k) = \frac{\binom{N_A}{k}\binom{N-N_A}{n-k}}{\binom{N}{n}}, \quad 0 \le k \le n$

$X \sim \text{Uniform}(a, b)$, PDF: $f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Normal}(\mu, \sigma^2)$, PDF: $f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$

$X \sim \text{Exponential}(\lambda)$, PDF: $f_X(t) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Gamma}(r, \lambda)$, PDF: $f_X(t) = \begin{cases} \frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$(X_1, \ldots, X_k) \sim \text{Multinomial}(n, k, p_1, \ldots, p_k)$,
joint PMF: $p(a_1, \ldots, a_k) = \binom{n}{a_1, \ldots, a_k} p_1^{a_1} \cdots p_k^{a_k}$ for $a_1, \ldots, a_k \ge 0$ with $a_1 + \cdots + a_k = n$.